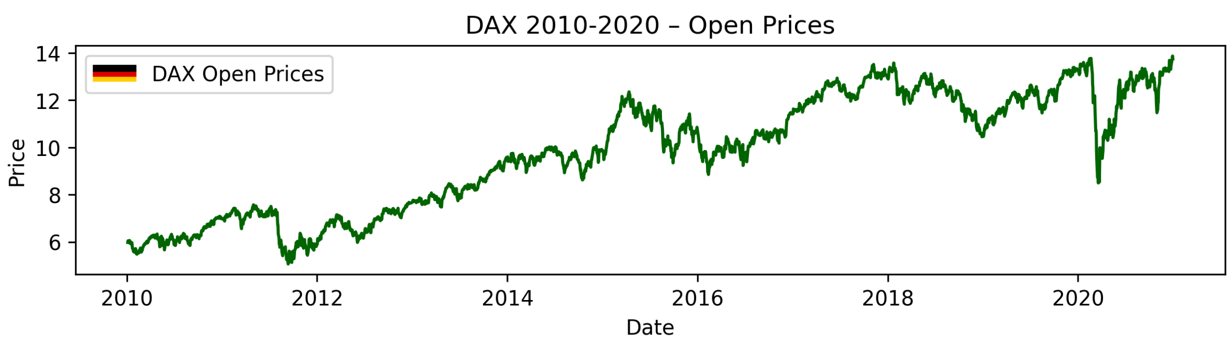
STATS 531 - Midterm Project

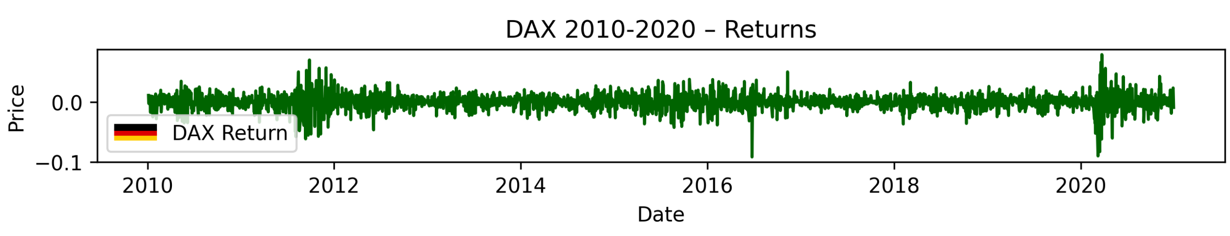
We look into the DAX Stock Index in Germany, and investigate how we can model the stock's opening prices, volatility, and other features. The DAX index tracks the performance of the 40 largest companies listed on the Frankfurt Stock Exchange (STOXX Ltd. 2026). They represent the diversified economy of Germany. The DAX is generally known as the German equity benchmark, surviving for over 30 years. What are the best ways to model changes in the stock's price and its returns? How do we model its volatility, and how accurately?

1 Exploratory Data Analysis

1.0.1 Introduction

We look at the daily stock price open for the DAX index for every day from 2010 to 2020 through a time series graph. The data above for stock open shows a roughly nonstationary data. The mean is not constant, and we have a large dip in 2020, associated with covid (Caporale and Gil-Alana 2022). Then we analyze volatility for the DAX stock, $y_n = log(z_n) - log(z_{n-1})$ (Ionides 2026).

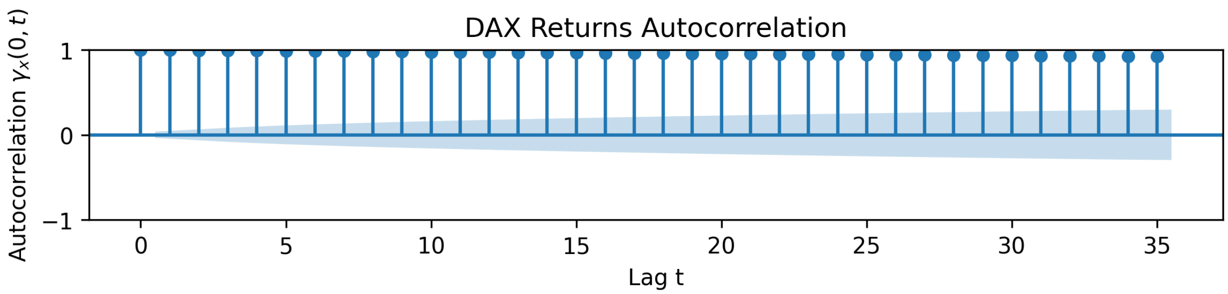




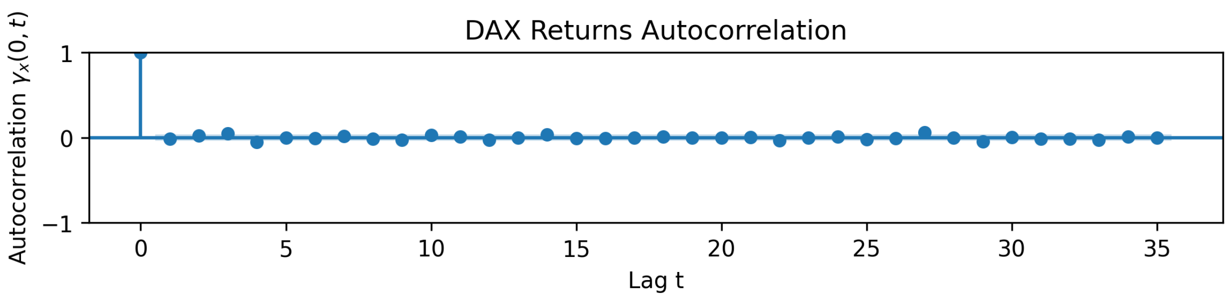
Through visual inspection, volatility is stationary with a more or less constant mean.

1.0.2 Autocorrelation

We look at the autocorrelation function below to see how much lags depend on each other. DAX opening prices are all highly correlated from the original value down until the 35th lag. Correlation then drops slowly. This tells us that a model that depends on its lags seems suitable.

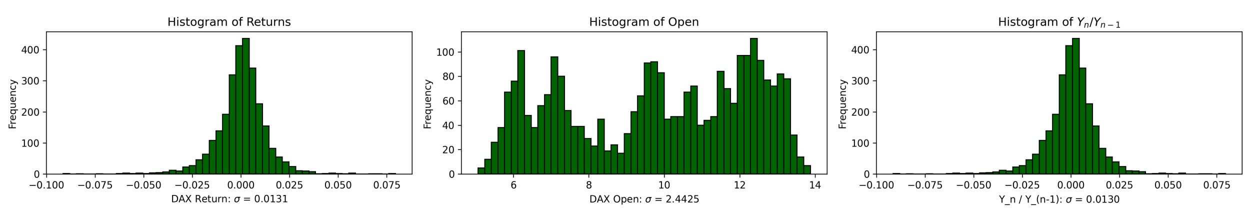


The autocorrelation graph for DAX Returns is below. The correlation of the returns are such that they could be i.i.d.. This means that the relationships between the first price return and its lags aren't as heavily correlated.



1.0.3 Distributions

Now we show the histogram for the opening prices, returns, and volatility. These display the shapes of our data. While returns and volatility are normally distributed, opening prices are more uniformally distributed.



1.0.4 Stationary vs Non Stationary

The time series graphs for DAX Opening prices look nonstationary through visual inspection, and the returns for DAX by day look stationary. We obtain empirical data to show whether the data is stationary or not through the Augmented Dickey Fuller Test. A time series model has a unit root if its first difference is stationary. For a linear time series model, this is equivalent to a $(1 - B)$

factor in the AR polynomial $\phi(B)$. The Augmented Dickey Fuller Test will look for evidence of a unit root. We use the test to verify the DAX Opening prices are nonstationary, and the returns for the DAX stock index are stationary.

$$
\begin{aligned}
&H_0 : \text{The process } \{x_t\} \text{ is nonstationary} \\
&\quad \exists\, t, h, k \text{ such that } (x_t, x_{t+1}, \dots, x_{t+k}) \overset{d}{\neq} (x_{t+h}, x_{t+1+h}, \dots, x_{t+k+h}) \\
&H_A : \text{The process } \{x_t\} \text{ is stationary} \\
&\quad \forall\, t, h, k,\ (x_t, x_{t+1}, \dots, x_{t+k}) \overset{d}{=} (x_{t+h}, x_{t+1+h}, \dots, x_{t+k+h})
\end{aligned}
$$

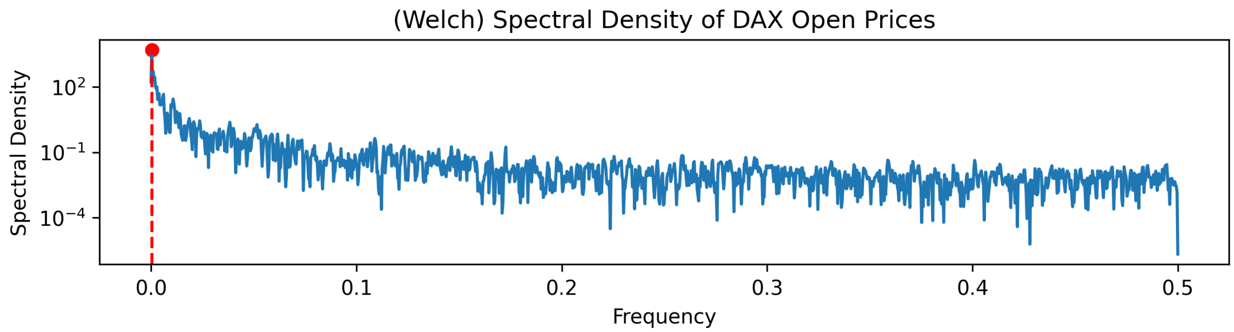
	0	1
Series	DAX Open	DAX Returns
ADF Statistic	-1.35435	-26.565984
p-value	0.603955	0.0
Lags Used	4	3
Observations	2785	2785
Critical Value (5%)	-2.862578	-2.862578
Reject the H0?	No	Yes

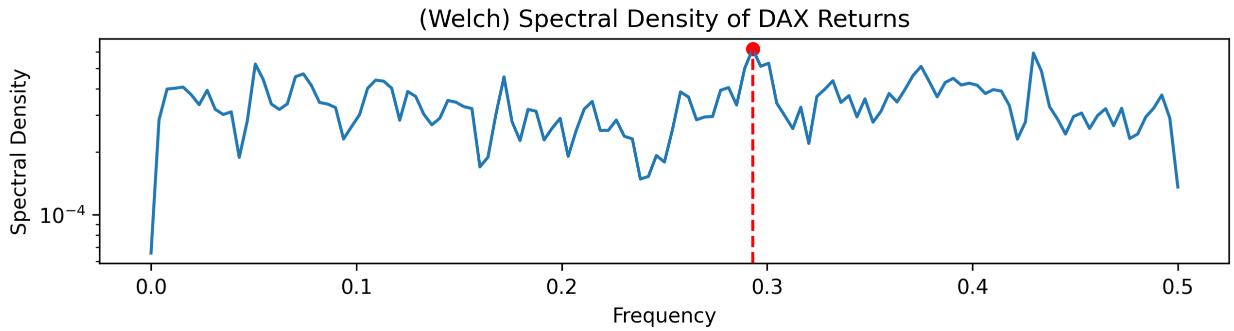
Therefore, the DAX Opens has a p value of 0.603955, so, we fail to reject the null, so we may assume it is nonstationary. On the other hand, the DAX Return has a p value of 0 after running the test. We reject the null and there is evidence to show that the data is stationary.

1.0.5 Spectral Density Function

An estimator of the spectral density function determine a signal's frequency context content from a time series sample. It is measured as $\lambda(\omega) = \int_{\infty}^{-\infty} \gamma(x) e^{-2\pi i \omega x}$ (Ionides 2026). In this section, we look at the frequencies for the DAX Stock Prices and Returns. It splits data into overlapping segments, computes modified periodograms, and averages them to reduce the variance. From the graphs, we may visually inspect the point at which the spectral density is the highest. This will be the dominant frequency, the most common rate at which an oscillation in the stock index happens.

Below, we show frequency vs spectral density function of the DAX Open and Return values.





The highest spectral densities in our graphs above are at frequencies of 0.00036 and 0.3, which indicates that these are the dominant frequencies for DAX open and return values, respectively. The periods are 2777 and 3.3 for the dominant oscillation. These make sense, as, we have roughly 2777 data, so, there is one unique non stationary pattern for open prices. Returns are stationary, and repeat roughly every 3.3 stock measurements, which is roughly 3.3 days.

2 Fitting Time Series Models

Let's train models on time series data to accurately reflect our observations on the DAX Stock Index from the exploratory data analysis, respecting stationarity.

2.0.1 ARMA and ARIMA

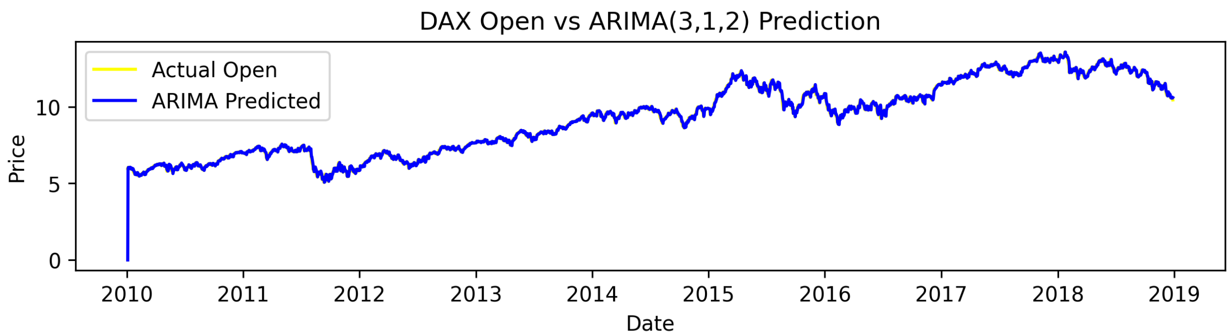
How do we forecast DAX Stock Prices and Returns? The exploratory data analysis reveals that these values depend on lags and differencing. This information is captured in an ARMA model (Shumway and Stoffer 2000).

$$y_n = \phi_1 y_{n-1} + \phi_2 y_{n-2} + \phi_3 y_{n-3} + \theta_1 \epsilon_n + \theta_2 \epsilon_{n-1}$$
$$y_n = \phi_1 B y_n + \phi_2 B^2 y_n + \phi_3 B^3 y_n + \theta_1 \epsilon_n + \theta_2 B \epsilon_n$$
$$y_n \phi(B) = \theta(B) \epsilon_n$$

2.0.1.1 DAX Open Prices

Let's analyze the results of an ARIMA model on the DAX Open Prices; our visual analysis and ADF test revealed that the DAX Open Prices as a time series are nonstationary. Because of the COVID wave in 2020, for purposes of fitting a model, we use all results from Januray 1, 2010 to December 31, 2018. The best ARIMA model for DAX Open Prices is an ARIMA(3,1,2). The shape matches, and has a relatively low AIC of -3916.7 - it is trained and graphed on top of raw data below, overlapping almost perfectly. Before measuring residuals, we compare models by $AIC = 2k - 2ln(\hat{L})$, which punishes complexity and lack of goodness of fit. A lower AIC score is better.

```
/Users/Ethan/Library/r-miniconda/envs/quarto/lib/python3.11/site-packages/statsmodels/tsa/state
  warn('Non-stationary starting autoregressive parameters'
/Users/Ethan/Library/r-miniconda/envs/quarto/lib/python3.11/site-packages/statsmodels/tsa/state
  warn('Non-invertible starting MA parameters found.'
```



The AIC values for the ARIMA models of different number of parameters (ex, ARIMA(1,0,1), ARIMA(2,0,2), etc.) are in the tables below.

Table 2: AIC for ARIMA(p,d,q) Model Comparison, as calculated from arma_aic_run.py

	MA Order				
AR Order	0	1	2	3	4
0	10629.7	7529.3	5005.4	3147.4	1892.3
1	-3897.5	-3895.6	-3893.6	-3893.5	-3892.2
2	-3895.6	-3894.0	-3892.1	-3891.9	-3891.6
3	-3893.6	-3892.4	-3891.1	-3898.6	-3896.7
4	-3893.6	-3891.3	-3889.2	-3896.6	-3895.1

(a) AIC for $d = 0$

	MA Order				
AR Order	0	1	2	3	4
0	-3905.2	-3903.3	-3901.3	-3901.2	-3899.9
1	-3903.3	-3901.3	-3899.3	-3899.3	-3901.8
2	-3901.3	-3899.8	-3916.6	-3906	-3903.5
3	-3901.3	-3899.3	-3916.7	-3904.1	-3903.7
4	-3899.8	-3901.5	-3903.9	-3902.4	-3915.5

(b) AIC for $d = 1$

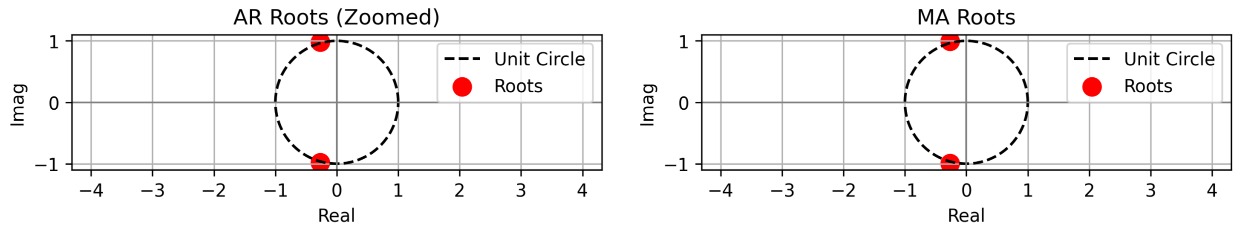
	MA Order				
AR Order	0	1	2	3	4
0	-2267.9	-3893.6	-3891.6	-3889.7	-3889.5
1	-2939.6	-3891.6	-3889.6	-3887.7	-3885.7
2	-3253.7	-3889.7	-3887.6	-3887.4	-3904.9
3	-3378.4	-3889.6	-3885.7	-3885.2	-3888.2
4	-3432	-3888.1	-3885.7	-3883.3	-3898.7

(c) AIC for $d = 2$

2.0.1.1.1 Model Diagnosis

We check causality and invertibility in this section. Causality is when the AR process has all roots that lie outside the unit circle (Shumway and Stoffer 2000). Invertability is when the MA process has all roots that lie outside the unit circle. We diagnose our best solution, ARIMA (3,1,2) below - it is both causal and invertible.

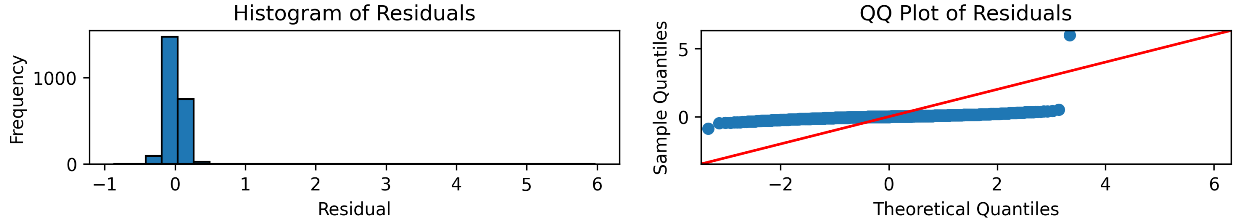
```
/Users/Ethan/Library/r-miniconda/envs/quarto/lib/python3.11/site-packages/statsmodels/tsa/state
  warn('Non-stationary starting autoregressive parameters'
/Users/Ethan/Library/r-miniconda/envs/quarto/lib/python3.11/site-packages/statsmodels/tsa/state
  warn('Non-invertible starting MA parameters found.'
```



2.0.1.1.2 Residual Analysis

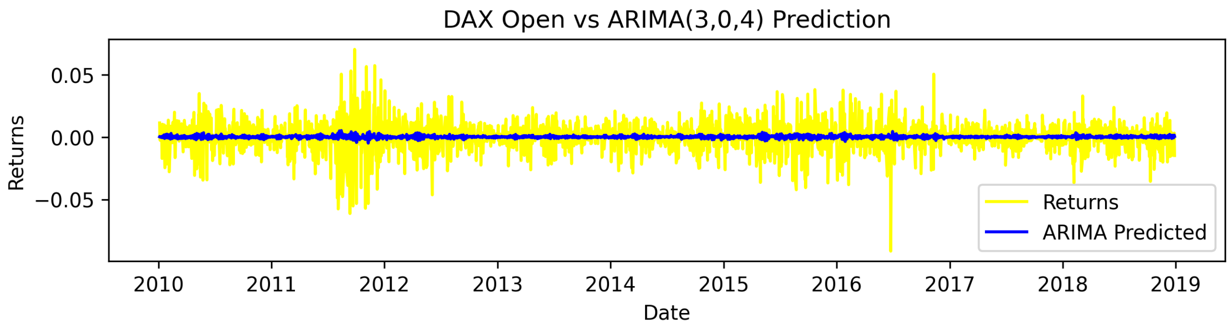
We look at the residuals. Both the qq plot and the histogram indicate that the residuals are normally distributed $\epsilon \approx N(0, \sigma^2)$. This indicates that the ARIMA(3,1,2) is a good fit.

```
/Users/Ethan/Library/r-miniconda/envs/quarto/lib/python3.11/site-packages/statsmodels/tsa/state
  warn('Non-stationary starting autoregressive parameters'
/Users/Ethan/Library/r-miniconda/envs/quarto/lib/python3.11/site-packages/statsmodels/tsa/state
  warn('Non-invertible starting MA parameters found.'
```



2.0.1.2 DAX Returns

Let's run the same procedure for the returns. The best model is ARIMA(3, 0, 4). The empirical and visual data shows that the returns are stationary, so, having do differencing component ($d = 0$) makes sense for this data.



Now, we look at AIC for every combination of parameters in ARIMA(p,d,q) for DAX returns.

Table 3: AIC for ARIMA(p,d,q) Model Comparison, as calculated from arma_aic_run.py

	MA Order				
AR Order	0	1	2	3	4
0	-14028.1	-14026.4	-14024.4	-14022.6	-14024.4
1	-14026.4	-14024.4	-14022.1	-14019.3	-14025.4
2	-14024.4	-14022.1	-14020.1	-14020.4	-14025.2
3	-14022.7	-14020.6	-14018.1	-14019.7	-14029.4
4	-14024	-14022.1	-14020.1	-14017.4	-14026

(a) AIC for $d = 0$

	MA Order				
AR Order	0	1	2	3	4
0	-12372.7	-14012.7	-14011.1	-14009.1	-14007.2
1	-13058.6	-14011.1	-14009.6	-14005.9	-14005.6
2	-13350.5	-14008.8	-14007.2	-14007.9	-14000.3
3	-13464.5	-14006.6	-14004.7	-14006.4	-14004.5
4	-13530.4	-14007.8	-13995.3	-13982	-13995.4

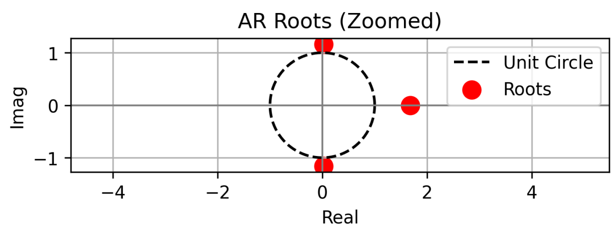
(b) AIC for $d = 1$

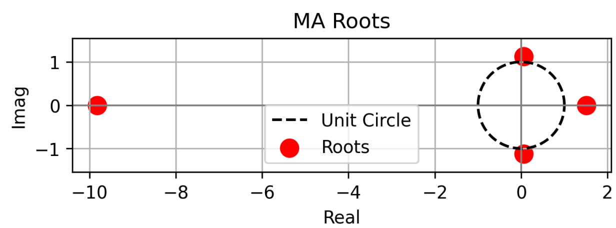
	MA Order				
AR Order	0	1	2	3	4
0	-9787.3	-12356.1	-1398	-13976.2	-13974.3
1	-11163.8	-13040.6	-13981.9	-13979.5	-13978.2
2	-11898.1	-13327.1	-13115.6	-13972.6	-13931.5
3	-12303.4	-13441.1	-13333.5	-13173.1	-13058.8
4	-12519.4	-13393.4	-13370.9	-13340.9	-13269.3

(c) AIC for $d = 2$

2.0.1.2.1 Model Diagnosis

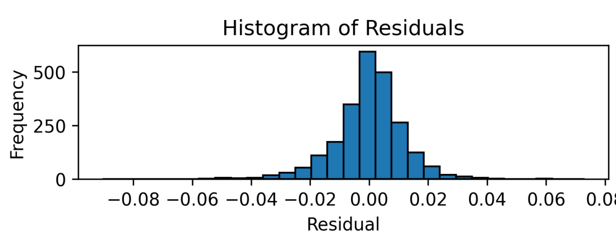
We assess causality and invertability. All solutions lie outside the unit circle. Therefore, our model is causal and invertible.

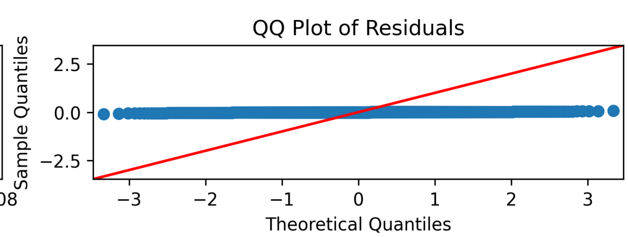




2.0.1.2.2 Residual Analysis

We look at the residuals. Both the qq plot and the histogram indicate that the residuals are normally distributed $\epsilon \approx N(0, \sigma^2)$. This indicates that the ARIMA(3,0,4) is a good fit.





2.0.2 SARMA and SARIMA

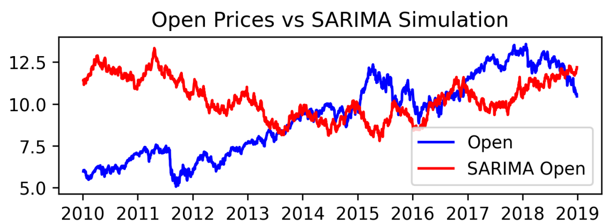
The Seasonal Autoregressive Moving Average Model (SARMA) adds a seasonal component to the ARMA and ARIMA Models. Instead of considering every lag from 1 to 35, for example, we can take lags 1, 12, and 13, repeatedly (Shumway and Stoffer 2000). This allows for monthly and yearly coefficients in our model, as one would expect with stock. We show a representation of SARIMAX(2,0,2)(1,0,0,12), and how well it performs below.

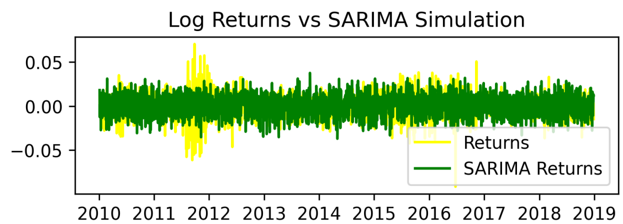
$$By_n = y_{n-1}; B^2 y_n = y_{n-2}; B^{12} y_n = y_{n-12}$$

$$y_t - \phi_1 y_{t-1} - \phi_2 y_{t-2} - \Phi_1 y_{t-12} + \Phi_1 \phi_1 y_{t-13} + \Phi_1 \phi_2 y_{t-14} = \epsilon_t + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2}$$

$$(1 - \Phi_1 B^{12})(1 - \phi_1 B - \phi_2 B^2)\, y_t = (1 + \theta_1 B + \theta_2 B^2)\, \epsilon_t$$

$$\Phi(B^{12})\, \phi(B)\, y_t = \theta(B)\, \Theta(B^{12})\, \epsilon_t$$





The SARIMA models for DAX Open prices and returns have AIC values of -3897.6 and -14029.4.

2.0.3 GARCH

The model selection above arguably did not sufficiently rule out a mean + white noise model for the returns data. For example, AIC favored ARMA(0, 0) as one of the top models, with the exception of models discussed above which were found to be numerically unstable. As such, we find it appropriate to shift the problem towards trying to predict the volatility of the returns, where numerical tools similar to those above can be extended to a problem where they are more likely to be successful. The GARCH(1, 1) models the following:

A GARCH model can model showing the volatility of the DAX Stock. The GARCH(1, 1) is shown below.

$$r_t = \mu_t + \varepsilon_t; \qquad \varepsilon_t = \sigma_t e_t; \qquad \sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$$

where r_t are the returns, μ_t is the mean function, ε_t is the residual at time t, σ_t^2 is the volatility at time t, and e_t is some iid noise distribution which we assume here is scaled to have variance 1 (say, $e_t \sim N(0,1)$ for simplicity).

We see that GARCH models essentially fit an ARMA model to σ_t^2 where the moving average terms are the squared residuals of the returns series. Thus, GARCH provides us an estimate of σ_t^2, which allows us to predict periods of high or low volatility in the market.

However, as we cannot observe volatility directly, we need a point of comparison for our GARCH estimate of σ_t^2. For simplicity, assume that $\mu_t = 0$ for all t. Indeed, for our log returns data, this seems appropriate, given that $ARMA(0,0)$ was favorable via AIC, there was constant trend visible in the plot, and since $\bar{x} \approx 0.02$ for our data. Then, as we show in the Supplementary Material, r_t^2 is a conditionally unbiased estimator for σ_t^2 (and the two are equal in marginal expectation). Thus, in that sense, it is theoretically favorable to use r_t^2 as a realized volatility proxy for comparison. The disadvantage is that it has high variance. One can also use "smoothed" proxy for volatility, which is the rolling average standard deviation (or variance) of the returns over some period of time T, where in this paper we use $T = 5$. This reduces the variance but increases bias. We mostly stick with r_t^2 here, as that seems to be favored in the literature and among practitioners for daily data [springer book citation , quantinsti citation].

2.0.3.1 Model Fitting

We first needed to find an appropriate order for our GARCH model. After performing a grid search, we found GARCH(1, 1) minimized the AIC and proceeded wi

We first needed to find an appropriate error distribution for the e_t term as in our notation above. Financial data can often have heavy tails, and errors here may not necessarily be symmetric. We performed a grid search for four different error distributions (Normal, t, Skew-t, and Generalized Error Distribution) and found that GARCH(1, 1) minimized the AIC in all four cases. Thus, we narrowed our search to GARCH(1, 1) and compared QQ plots of the standardized residuals obtained from fitting GARCH(1, 1) to log returns from 2010-2018 for each error distribution. We found that the Skew-t distribution had the best QQ plot, shown in Figure 2, indicating that the error terms e_t most resembled the skew-t distribution for our dataset. Figure 3 compares the GARCH estimate of volatility to the smoothed estimate. Thus, we see that the model was generally able to adapt to the volatility, following the highs and lows, though its estimates were a more tamed version of the realized proxy. However, using the unbiased realized volatility in Figure 4 shows how large errors can dominate the dataset and skew model performance. Metrics from in-sample performance the 2010-2018 data are shown below (estimated volatility and proxies are for variance):

Table 4: Model error comparison

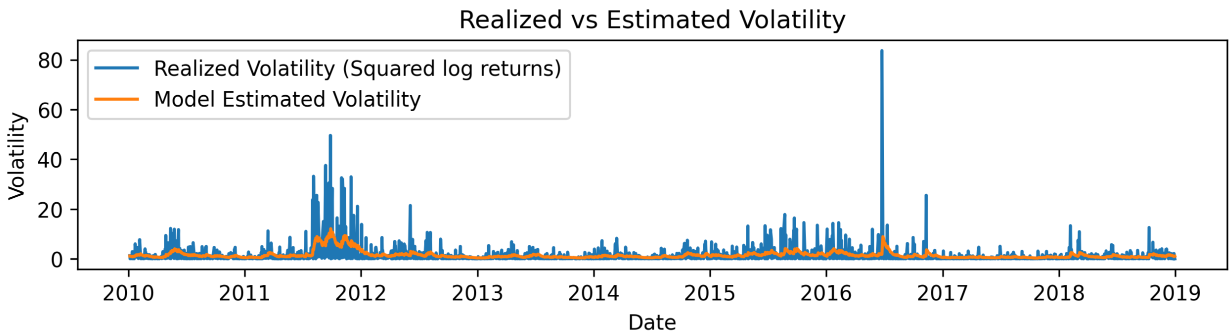
Metric	Unbiased Proxy	Smoothed Proxy
Mean Absolute Error	1.629	0.784
Mean Squared Error	12.420	2.266

To better assess the performance of this model, we used a rolling window forecast as suggested in [citation1, citation2], in which we used the previous 2000 days to forecast the volatility of the next day, and repeat for each day in 2019. Comparison with the smoothed estimate is shown in Figure 1. Our forecast metrics are shown below.

Table 5: Model error comparison

Metric	Unbiased Proxy	Smoothed Proxy
Mean Absolute Error	0.874	0.432

Metric	Unbiased Proxy	Smoothed Proxy
Mean Squared Error	1.522	0.253



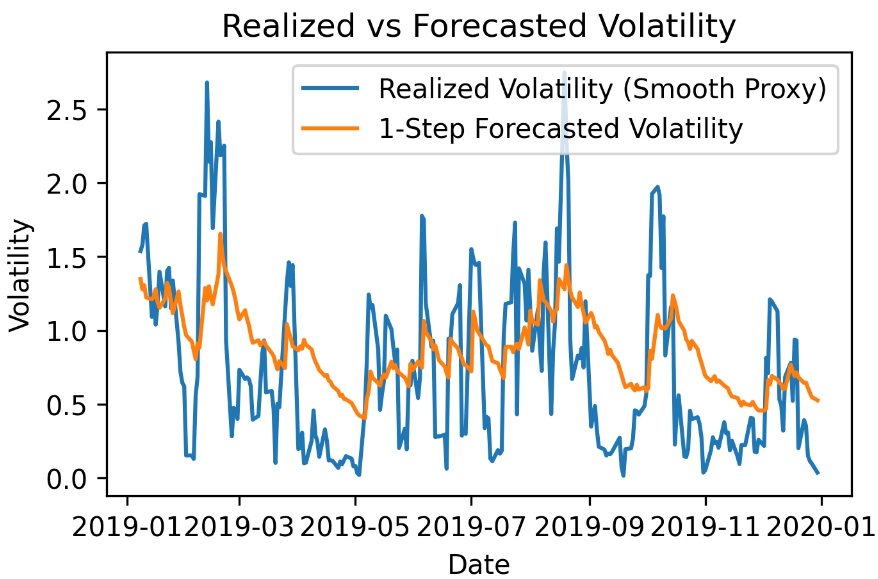

```
Iteration:      1,   Func. Count:      7,   Neg. LLF: 26778.02804605342
Iteration:      2,   Func. Count:     14,   Neg. LLF: 3662.428395356824
Iteration:      3,   Func. Count:     21,   Neg. LLF: 5986.833296937845
Iteration:      4,   Func. Count:     28,   Neg. LLF: 44773.3566274717
Iteration:      5,   Func. Count:     36,   Neg. LLF: 5423.164544157857
Iteration:      6,   Func. Count:     43,   Neg. LLF: 6285.875048266808
Iteration:      7,   Func. Count:     50,   Neg. LLF: 3440.942446827034
Iteration:      8,   Func. Count:     57,   Neg. LLF: 5399.781202915614
Iteration:      9,   Func. Count:     66,   Neg. LLF: 3439.6899451141153
Iteration:     10,   Func. Count:     73,   Neg. LLF: 3439.6818256797515
Iteration:     11,   Func. Count:     80,   Neg. LLF: 3439.6802622495165
Iteration:     12,   Func. Count:     85,   Neg. LLF: 3439.680262249554
Optimization terminated successfully    (Exit mode 0)
            Current function value: 3439.6802622495165
            Iterations: 12
            Function evaluations: 85
            Gradient evaluations: 12
```

3 Connection to Previous Projects

As with several other previous STATS 531 projects [example 1, example 2, example 3], we chose to study financial time series data. One of our primary goals, however, was to try and extend the work of [mainexample], which also tried to model volatility in stock returns. However, we tried to overcome some of the limitations in their study by first, not using GARCH to analyze volatility of volatility, as, though that may be statistically interesting, is most likely misspecified by GARCH, as the reviewer pointed out [reviewcitation]. Next, [mainexample] was limited in part by the fact that their QQ residual plots throughout their paper indicated misspecification due to heavy tails. They indicated that they attempted to use t-distributed errors but were unable to due to failures

<Figure size 1650x1050 with 0 Axes>

(a) Comparison of Smoothed Volatility to Estimated Volatility



(b)

Figure 1

in "model explanation and diagnosis". By identifying and implementing an error distribution that fails less drastically at the tails using the skewed t distribution, as shown in the QQ plot for our implementation Figure 2, we were able to at least partially overcome a key limitation in their analysis.

4 Supplementary Material

Now, for simplicity, assume that $\mu_t = 0$ for all t. Indeed, for our log returns data, this seems appropriate, given that $ARMA(0,0)$ was favorable via AIC, there was constant trend visible in the plot, and since $\bar{x} \approx 0.02$ for our data. Then we have $E(r_t^2 \mid \mathcal{F}_{t-1}) = Var(r_t \mid \mathcal{F}_{t-1}) = \sigma_t^2$ and hence $E(r_t^2) = E(E(r_t^2 \mid \mathcal{F}_{t-1})) = E(\sigma_t^2)$. Thus, r_t^2 is a conditionally unbiased estimator for σ_t^2 (and the two are equal in marginal expectation). Thus, it is favorable to use r_t^2 as a volatility proxy.

4.1 Proof that $\sigma_t^2 = Var(r_t - \mu_t \mid \mathcal{F}_{t-1})$

If we denote $\mathcal{F}_{t-1}$ as the σ-algebra (intuitively, information) generated by $\varepsilon_t, \sigma_{t-1}^2$ then we have

$$\begin{aligned} Var(r_t - \mu_t \mid \mathcal{F}_{t-1}) = Var(\varepsilon_t \mid \mathcal{F}_{t-1}) &= Var(\sigma_t e_t \mid \mathcal{F}_{t-1}) \\ &= \sigma_t^2 Var(e_t \mid \mathcal{F}_{t-1}) \qquad (\sigma_t \text{ is } \mathcal{F}_{t-1} \text{ measurable}) \\ &= \sigma_t^2 Var(e_t) = \sigma_t^2 \qquad (e_t \text{ is independent of } \mathcal{F}_{t-1}) \end{aligned}$$

4.2 Proof that squared returns are conditionally unbiased

We have

$$E(r_t^2 \mid \mathcal{F}_{t-1}) = Var(r_t \mid \mathcal{F}_{t-1}) = \sigma_t^2,$$

and hence

$$E(r_t^2) = E(E(r_t^2 \mid \mathcal{F}_{t-1})) = E(\sigma_t^2),$$

4.3 Supplementary GARCH plots

4.3.1 Conclusion

In conclusion, we show that modeling changes in the DAX Stock Open and Returns can be achieved through ARMA models with low AIC and mean zero residuals - these metrics indicate low complexity, goodness of fit, and low errors. We therefore solve the problem of forecasting DAX Stock prices. We expand upon it by giving insights into causality and invertibility, and we demonstrate modeling volatility with GARCH.

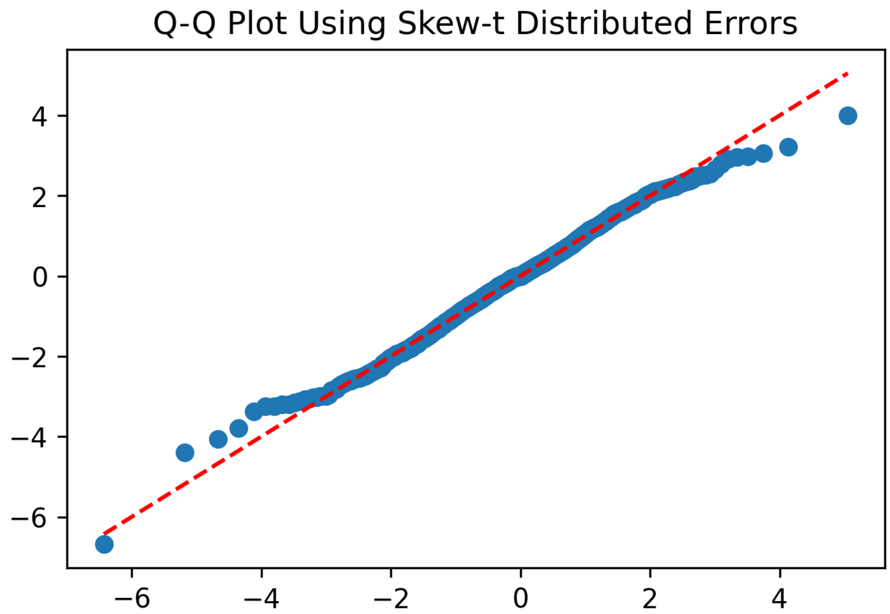


Figure 2: QQ Plot of Standardized Residuals

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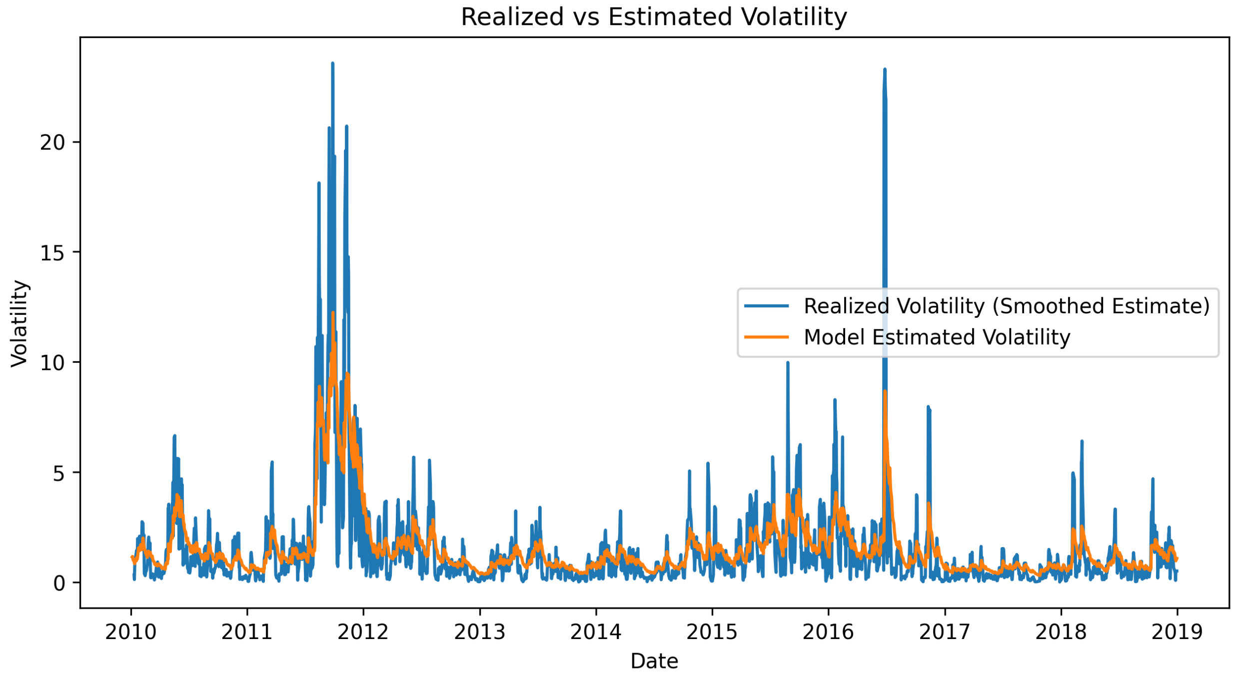
(a) Comparison of Smoothed Volatility to Estimated Volatility

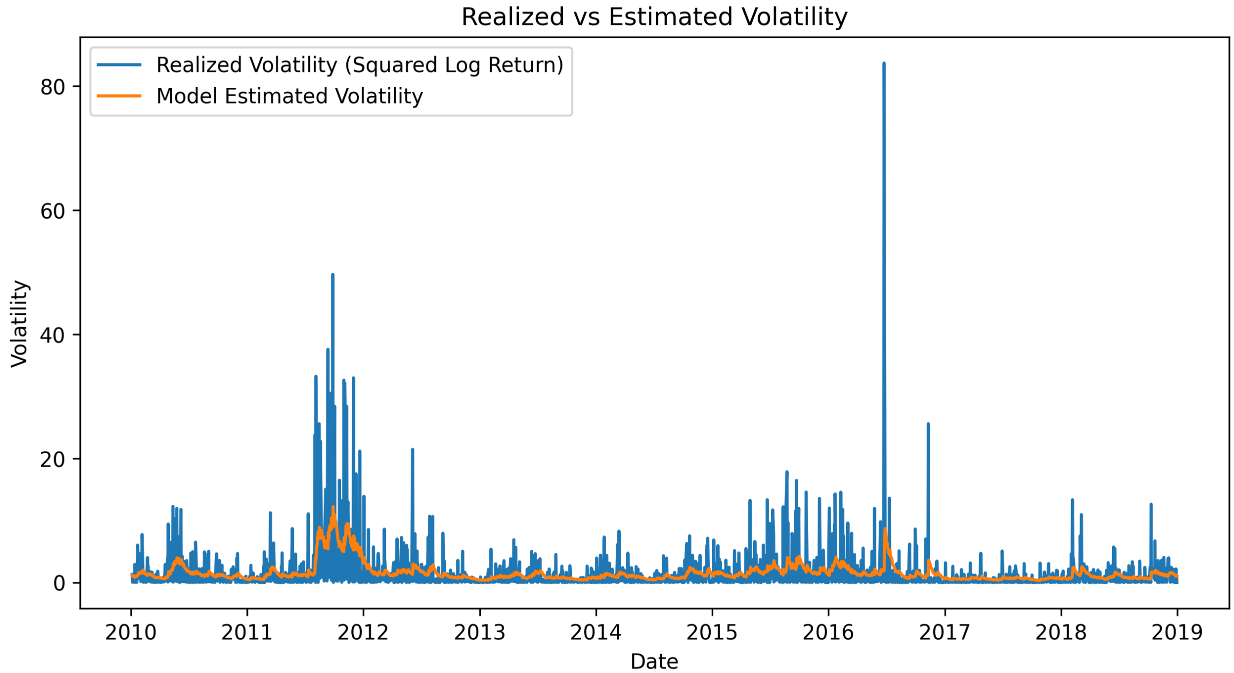


(b)

Figure 3

<Figure size 1650x1050 with 0 Axes>

(a) Comparison of Realized Volatility to Estimated Volatility



(b)

Figure 4